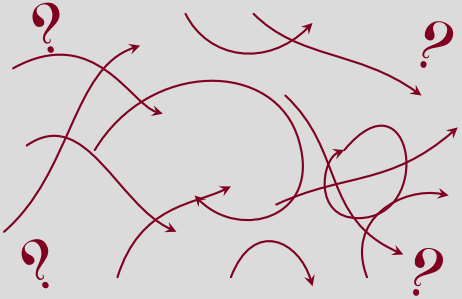
Which way is up?



Andrew Fowlie

Introduction to Vector Calculus

CC BY 4.0 Attribution 4.0 International

Contents

1. Vector Fields

2. Gradient vector field

3. Which way is up?

4. Modern relevance

Section 1

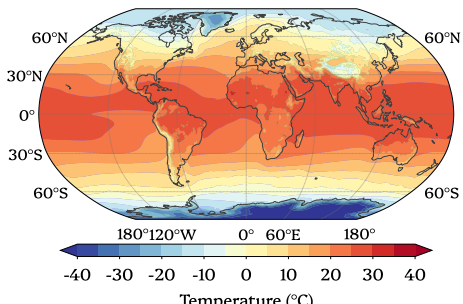
Vector Fields

Fields

- We can think of a field as a function of position in space
- That space could be a two-dimensional plane, $\mathbb{R}^2$; the three-dimensional space around us, $\mathbb{R}^3$; or a higher-dimensional, abstract vector space, $\mathbb{R}^n$
- The field could be a **scalar** field, a **vector** field or even a **tensor** field
- Today we'll consider scalar and vector fields in two dimensions

Scalar Fields

- Scalar fields are simple — they assign a scalar, i.e., a single real number, to every position in the space
- For example, a function that assigns a temperature to every position on Earth is a scalar field



Scalar Fields

- Scalar fields are simple — they assign a scalar, i.e., a single real number, to every position in the space
- For example, a function that assigns a temperature to every position on Earth is a scalar field

Definition | Scalar field — Let D be a region in $\mathbb{R}^2$. A two-dimensional *scalar field* is a function, f, that assigns a scalar $f(x, y)$ to each point (x, y) in D.

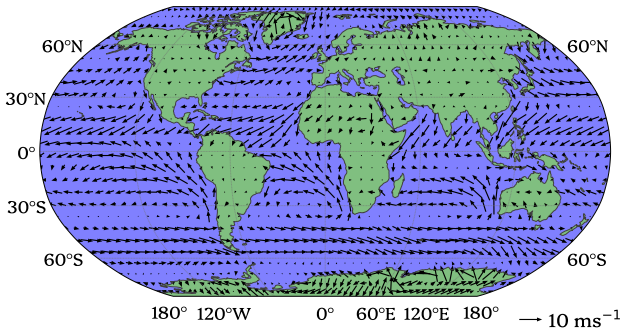
Vector fields

- We now consider a *vector field* — this time, we assign a vector to every position in the space
- If our space is two-dimensional, we assign a two-dimensional vector; if our space is n-dimensional, we assign an n-dimensional vector
- For example, a function that assigns a horizontal wind velocity vector to every position on Earth is a two-dimensional vector field

Definition | Vector field — Let D be a region in $\mathbb{R}^2$. A two-dimensional *vector field* is a function, $\mathbf{f}$, that assigns a two-dimensional vector $\mathbf{f}(x, y)$ to each point (x, y) in D.

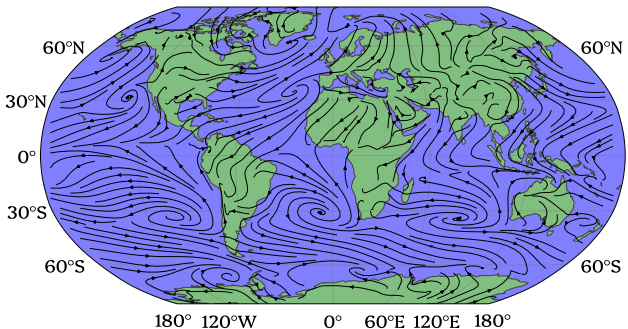
Quiver plots

- What do vector fields **look** like? How can we visualise them?
- In a *quiver plot*, the length and direction of the arrow shows the vector field magnitude and direction



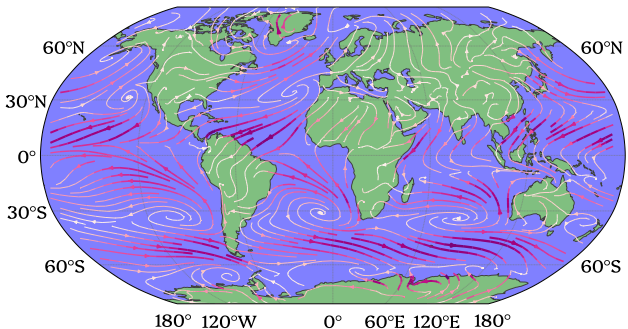
Field lines

- Quiver plots can be hard to read. Instead, we can plot *field lines*
- The field lines are always tangent to the vector field
- Easy to see direction, but hard to see magnitude



Field lines

- Quiver plots can be hard to read. Instead, we can plot *field lines*
- The field lines are always tangent to the vector field
- Magnitude can be shown by line thickness and colour, or sometimes density



Example of a vector field

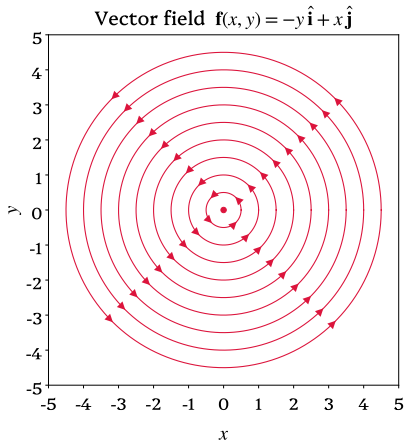
- Consider the two-dimensional vector field

$$\mathbf{f}(x, y) = -y\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}$$

- The vectors $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are unit vectors in the x- and y-directions
- Can you guess what the field lines look like?

Circular

- The field $\mathbf{f}(x, y) \cdot \mathbf{r} = -yx + xy = 0$
- Thus the field is perpendicular to radial vectors
- The tangent to a circle is perpendicular to its radius
- The field lines are circles about the origin



Section 2

Gradient vector field

Gradient vector field

- How does a scalar field change over the space?
- We can compute the partial derivatives

$$\frac{\partial f(x,y)}{\partial x} \quad \text{and} \quad \frac{\partial f(x,y)}{\partial y}$$

- Remember, partial derivatives measure change in one direction, holding the other coordinates fixed
- We can arrange the partial derivatives into a vector field

$$\nabla f(x,y) = \frac{\partial f(x,y)}{\partial x}\,\hat{\mathbf{i}} + \frac{\partial f(x,y)}{\partial y}\,\hat{\mathbf{j}}$$

- We read ∇ as *grad* or *del*

What is that ∇ thing?

Definition | **Gradient** — The gradient operator $\boldsymbol{\nabla}$ turns a scalar field $f(x, y)$ into a vector field containing its partial derivatives

In Cartesian coordinates, we can write it as

$$\boldsymbol{\nabla} f(x, y) = \frac{\partial f(x, y)}{\partial x}\,\hat{\mathbf{i}} + \frac{\partial f(x, y)}{\partial y}\,\hat{\mathbf{j}}$$

You'll see in later lectures that we can write it in other coordinate systems, such as spherical and cylindrical coordinates

An example of that ∇ thing

- Take, for example, a scalar field

$$f(x, y) = \frac{1}{2}\left(x^2 + y^2\right)$$

- We compute the gradient field by first computing the partial derivatives

$$\frac{\partial f(x, y)}{\partial x} = x \quad \text{and} \quad \frac{\partial f(x, y)}{\partial y} = y$$

- Thus the gradient field

$$\nabla f(x, y) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$$

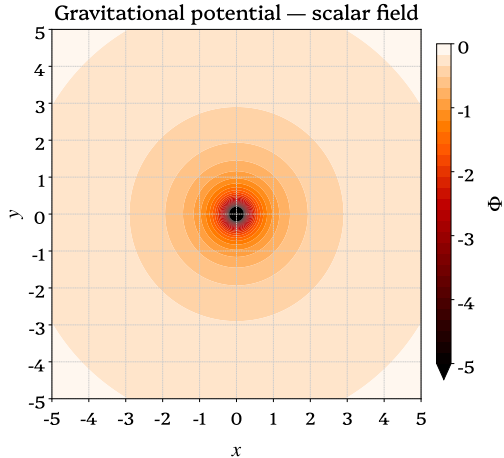
- What do the field lines look like?

Gravitational field

- Consider the gravitational potential around a point mass

$$\Phi(\mathbf{r}) = -\frac{GM}{r}$$

- G and M are physical constants and the radius $r = |\mathbf{r}| = \sqrt{x^2 + y^2}$
- The potential $\Phi(\mathbf{r})$ is a scalar field — it assigns a potential to every position in space



Gravitational field

- What is the negative gradient of this scalar field?

$$\mathbf{g}(\mathbf{r}) = -\nabla\Phi(\mathbf{r})$$

- The negative gradient $\mathbf{g}(\mathbf{r})$ is a vector field — it assigns a vector to every position in space
- Start by computing the partials

$$\frac{\partial\Phi(\mathbf{r})}{\partial x} = GM\frac{x}{r^3} \quad \text{and} \quad \frac{\partial\Phi(\mathbf{r})}{\partial y} = GM\frac{y}{r^3}$$

- Such that

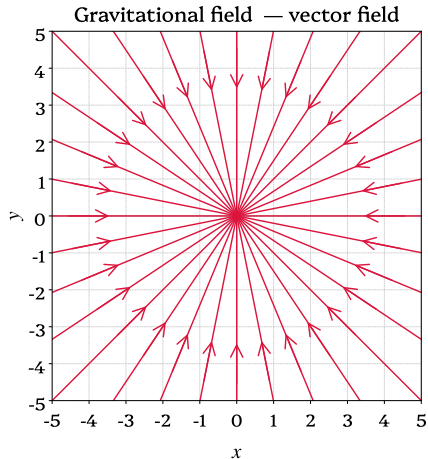
$$\mathbf{g}(\mathbf{r}) = -\frac{GM(x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}})}{r^3} = -\frac{GM}{r^2}\,\hat{\mathbf{r}}$$

Gravitational field

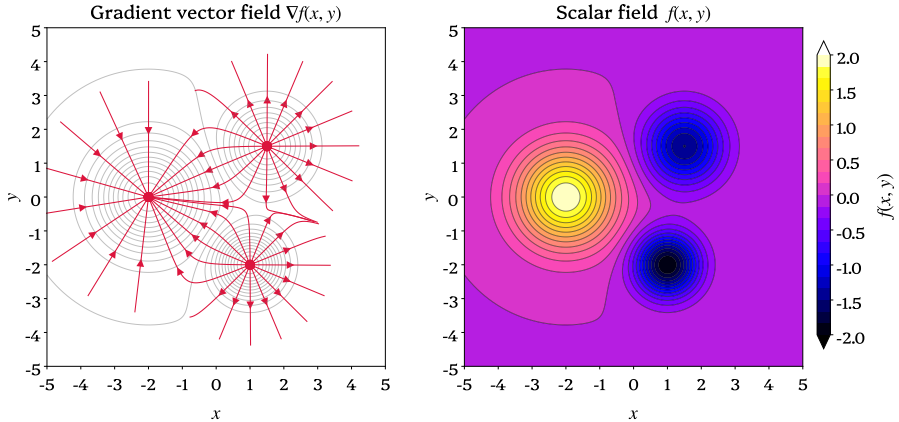
- Physically, $\mathbf{g}(\mathbf{r})$ can be interpreted as the gravitational **field**
- An object of mass m experiences a force

$$\mathbf{F}(\mathbf{r}) = m\,\mathbf{g}(\mathbf{r})$$

- The gradient links the gravitational potential to the gravitational field
- The negative gradient points towards decreasing values of the scalar field



Gradient field with multiple extrema



The gradient vectors: **1.** point in the direction of increasing scalar field; **2.** are always perpendicular to the contours of constant scalar field

Are vector fields always gradient vector fields?

- For a vector field $\mathbf{f}(x, y) = \nabla g(x, y)$, the scalar field $g(x, y)$ is called a scalar potential
- Can all vector fields be written as gradient vector fields?
- **No.** This is a special property. If $\mathbf{f}(x, y) = \nabla g(x, y)$, then

$$\frac{\partial}{\partial x}\mathbf{f}_y(x, y) = \frac{\partial}{\partial x}\frac{\partial}{\partial y}g(x, y) = \frac{\partial}{\partial y}\frac{\partial}{\partial x}g(x, y) = \frac{\partial}{\partial y}\mathbf{f}_x(x, y)$$

- We used the fact that we can swap the order of partial derivatives

Counterexample

- Consider again our vector field

$$\mathbf{f}(x, y) = -y\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}$$

- Does it satisfy our condition that $\partial_x \mathbf{f}_y(x, y) = \partial_y \mathbf{f}_x(x, y)$?

$$\partial_x \mathbf{f}_y(x, y) = 1 \quad \text{and} \quad \partial_y \mathbf{f}_x(x, y) = -1$$

- Thus it does not and **cannot** be written as a gradient vector field
- This field was purely **rotational** — the field lines were circular
- Gradient vector fields must be **irrotational** — we will define this formally next week when we discuss ***curl***

Properties

Here f and g are scalar fields and k is a scalar constant

Properties | Gradients

1. Sum rule: $\nabla(f+g) = \nabla f + \nabla g$
2. Difference rule: $\nabla(f-g) = \nabla f - \nabla g$
3. Multiple rule: $\nabla(kf) = k\nabla f$
4. Product rule: $\nabla(fg) = f\nabla g + g\nabla f$

Fundamental gradients

These may be proven by writing the gradient and vectors in component form, using $\mathbf{r} = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$. Here $\mathbf{a}$ is constant

Results | Gradients

1. $\nabla(\mathbf{a}\cdot\mathbf{r}) = \mathbf{a}$
2. $\nabla|\mathbf{r}-\mathbf{a}| = \dfrac{\mathbf{r}-\mathbf{a}}{|\mathbf{r}-\mathbf{a}|}$ *for* $\mathbf{r} \neq \mathbf{a}$
3. $\nabla r = \hat{\mathbf{r}}$ *for* $\mathbf{r} \neq 0$
4. $\nabla\dfrac{1}{r} = -\dfrac{\hat{\mathbf{r}}}{r^2}$ *for* $\mathbf{r} \neq 0$ — *we used this for the gravitational field*
5. $\nabla f(r) = \dfrac{df(r)}{dr}\,\hat{\mathbf{r}}$ — *radial fields have radial gradients*

Directional derivative

- We can measure how a scalar field changes in the x- and y-directions using partial derivatives
- What about an arbitrary direction, $\hat{\mathbf{u}}$?
- The **directional derivative** measures the rate of change of f in the direction $\hat{\mathbf{u}}$
- The directional derivative may be found from the gradient vector field

$$D_{\hat{\mathbf{u}}} f(x, y) = \nabla f(x, y) \cdot \hat{\mathbf{u}}$$

- So, which direction gives the **largest** directional derivative? **Which way is up?**

Section 3

Which way is up?

Which way is up?

- I am walking on a scalar field and I want to go uphill as quickly as possible
- **Which direction should I walk?**

Which way is up?

- I am walking on a scalar field and I want to go uphill as quickly as possible
- **Which direction should I walk?**
- Write the directional derivative using the angle between $\hat{\mathbf{u}}$ and the gradient vector field

$$D_{\hat{\mathbf{u}}} f(x, y) = |\nabla f(x, y)| \cos\theta$$

- For $|\nabla f(x, y)| \neq 0$, this has a maximum at $\cos\theta = 1$ and $\theta = 0$ and thus

$$\hat{\mathbf{u}} = \frac{\nabla f(x, y)}{|\nabla f(x, y)|}$$

- The gradient vector field points in the direction that maximises the directional derivative of f; **it points up, so this is the way I should walk**

Walking around the hill

- We've found up. Now I want to walk in a direction that doesn't gain or lose height
- **Which direction should I walk?**

Walking around the hill

- We've found up. Now I want to walk in a direction that doesn't gain or lose height
- **Which direction should I walk?**
- We need to find $\hat{u}$ such that

$$D_{\hat{\mathbf{u}}} f(x, y) = \nabla f(x, y) \cdot \hat{\mathbf{u}} = 0$$

- Similar to before, we can write this as

$$D_{\hat{\mathbf{u}}} f(x, y) = |\nabla f(x, y)| \cos\theta = 0$$

- This occurs when $\theta = \pi/2$ and thus $\hat{u}$ is perpendicular to $\nabla f(x, y)$
- These directions are tangent to **level curves** and **perpendicular** to the field lines of the gradient vector field
- **I should walk along these tangents**

Section 4

Modern relevance

Gradient descent

- We often want to **optimize** a multivariate function — find the arguments that maximize or minimize the function
- How can we find the minimum? **Follow the steepest path downwards!**
- Take steps in the direction $-\nabla f(x, y)$
- This is known as **gradient descent**

Machine Learning

- Neural networks contain many **parameters**, $\mathbf{p}$
- We choose them by minimizing a loss function, L, that compares the network output to the training data
- Gradient descent is a **cornerstone** of optimization and training in ML. With learning rate η

$$\mathbf{p}_{n+1} = \mathbf{p}_n - \eta \nabla L(\mathbf{p}_n \,|\, \text{Training data})$$

- In fact, ML often uses **stochastic** gradient descent — computes a noisy estimate of the gradient from a random **minibatch** of data

$$\mathbf{p}_{n+1} = \mathbf{p}_n - \eta \nabla L(\mathbf{p}_n \,|\, \text{Random minibatch of training data})$$

Further reading

- Introductory, thorough and clear — George B Thomas, Maurice D Weir, and Joel Hass. *Thomas' Calculus: Multivariable*. Addison-Wesley, 2010 and James Stewart, Daniel K Clegg, and Saleem Watson. *Multivariable calculus*. Cengage Learning, 2020
- Slightly more advanced, but beautiful and connects to physics and engineering — Fabian Waleffe. *Vector and Complex Calculus: A Textbook for Students of the Physical Sciences*. Oxford University Press, 2024

Summary

- **Gradient field?** A gradient field, ∇f, tells us how a scalar field changes in space
- **Which way is up?** ∇f points up, that is, in the direction of steepest increase of f
- **Which way is down?** $-\nabla f$ points down, that is, in the direction of steepest decrease of f
- **Which way nothing changes?** Directions perpendicular to ∇f
- **How is this useful?** Minimize e.g., a loss function using gradient descent when training neural networks